Viva Questions:

1. Define feedback

Feedback is the process of taking a fraction of output signal and feeding it back to the input of an electronic circuit.

2. Define positive feedback

Positive feedback in which a fraction of output signal is fed in phase with the input signal.

3. Define negative feedback

Negative feedback in which a fraction of output signal is fed out of phase with the input signal

4. Define sensitivity

Sensitivity is the ratio of percentage change in gain to the percentage change in circuit parameter.

5. What are the types of feedback?

*. Positive feedback.

*. Negative feedback.

6. Give an example for voltage-series feedback.

Non-inverting operational amplifier (voltage amplifier) is an example of voltage-series feedback.

7. Give the properties of negative feedback.

*. Reduce gain *. Reduces distortion *. Increases bandwidth *. Increase input impedance

*. Improves stability *. Reduces noise *. Improves Linearity. *. Decrease output impedance

8. Define voltages hunt feedback.

Voltage-shunt feedback is a feedback in which the output voltage is sampled and a fraction of it is fed back in parallel with the input

9. Define voltage series feedback.

Voltage-series feedback is a feedback in which the output voltage is sampled and a fraction of it is fed back in series with the input

10. Define current shunt feedback.

current shunt feedback is a feedback in which the output current is sampled and a fraction of it is fed back in parallel with the input.

Viva Questions:

1. What are power amplifiers?

Power amplifiers are electronics amplifiers designed to increasing the power level of a signal and deliver maximum power to load.

2. What are the types of power amplifiers?

* . class A power amplifier
* . class AB power amplifier
* . class B power amplifier
* . class C power amplifier

3. Define Conduction angle.

conduction angle is the angle of the input cycle during which the transistor conducts current.

4. Compare various power amplifier circuits.

class A : conduction angle 360°, low efficiency and low distortion
class B : conduction angle 180°, high efficiency and crossover distortion
class AB: conduction angle 180°-360°, moderate efficiency and low distortion
class C : conduction angle less than 180°, very high efficiency, high distortion.

5. Compare rectifier and regulator.

Rectifier : converts Ac to pulsating DC.

Regulator : Maintains a constant DC output voltage despite variations in input or load.

6. List the various features of power amplifier circuits.

* . High power output
* . High efficiency
* . Drive low-resistance loads
* . Large signal operation
* . Low output distortion
* . Heat dissipation is significant
* . High current handling capability

7. What is cross over distortion?

cross-over distortion is the distortion that occurs near the zero-crossing point of the output signal in a class B push-pull amplifier.

8. Why Class C power amplifier is commonly used?

class C power amplifiers are commonly used because they have very high efficiency and are mainly suitable for RF applications.

9. What are the types of Class B power amplifiers?

⇒ Transformer-coupled push-pull amplifier.

⇒ complementary-Symmetry push-pull amplifier.

10. What is frequency multiplier?

Frequency multiplier is an electronic circuit that generates an output signal whose frequency is an integer multiple of the input frequency.

Viva Questions:-

1. What is an Oscillator ?

An oscillator is a circuit that generates periodic signals without an external input.

2. What are the different types of oscillators?

⇒ RC oscillator ⇒ crystal oscillator

⇒ LC oscillator ⇒ Relaxation oscillator.

3. What are the conditions for oscillations?

The conditions for oscillations:

* Loop gain $A_v\beta = 1$
* Total phase shift is 0° or 360°.

4. Define frequency oscillation.

Frequency of oscillation is the number of cycles completed per second by an oscillator. unit: Hertz(Hz)

5. What is the application of RC phase shift oscillator?

The application of RC phase shift oscillators are:

⇒ used for audio frequency signal generation.

⇒ Function generators.

⇒ Electronic testing circuits.

Viva Questions:

1. What is an Oscillator circuit?

An Oscillator circuit is an electronic circuit that generates continuous periodic signals without an external input.

2. What are the different types of oscillators?

⇒ RC Oscillator ⇒ Crystal oscillator.

⇒ LC oscillator ⇒ Relaxation oscillator.

3. What are the conditions for oscillations?

According to Barkhausen conditions,

⇒ Loop gain $A_V\beta = 1$

⇒ Total phase shift must be 0° or 360°

4. Define frequency oscillation.

Frequency of oscillation is the number of cycles completed per second by an oscillator.

unit: Hertz (Hz)

5. What is the application of RF phase shift oscillator?

⇒ used for audio-frequency signal generation.

⇒ Function generators

⇒ Electronic testing circuits.

6. Define Oscillation.

Oscillation is the repetitive variation of a quantity about a mean value

7. What is a tuned amplifier?

A tuned amplifier is an amplifier that amplifies a specific range of frequencies using a tuned circuit.

8. What happens to the circuit above and below resonance?

Below resonance, the circuit behaves capacitively, while above resonance, the circuit behaves inductively.

9. What are the different coil losses?

The different coil losses are copper loss, hysteresis loss, eddy current loss and dielectric loss.

10. What is Q factor?

Q-factor is the ratio of resonant frequency to bandwidth of a tuned circuit.

$$Q = \frac{f_r}{BW}$$

Viva Questions:

1. What is Biasing?

Biasing is the process of applying DC Voltage or current to set the operating point of a Transistor.

2. What is CMRR?

CMMR is the ability of an amplifier to reject common-mode signals while amplifying the different signal. CMRR = A_d/A_c

3. What is an operational amplifier?

An operational amplifier is a high-gain differential amplifier used to amplify and process electrical signals.

4. What is open loop and close loop opamp?

An open-loop op amp operates without feedback, while closed-loop opamp operates with feedback.

5. Why practical opamp is not ideal?

A practical op-amp is not ideal because it has finite gain, input resistance and bandwidth, unlike an ideal op-Amp

6. Define input bias current (or) .Why do we use Rcomp resistor?

Input bias current is the average current flowing into the two input terminals of an op-Amp, and Rcomp is used to reduce the resulting output offset voltage.

7. Define input offset current.

Input offset current is the difference between the bias currents flowing into the two input terminals of an op-Amp.

8. Define input offset voltage.

Input offset voltage is the differential voltage required between the input terminals to make the output voltage zero.

9. What is inverting amplifier and where it is used mostly?

An inverting amplifier produces an output 180° out of phase with the input and is mostly used for signal amplification and phase inversion.

10. What is non inverting amplifier and its applications.

A non-inverting amplifier amplifies the input signal without changing its phase and is mainly used in signal amplification and voltage buffering.

VivaQuestions:-

1. Define an integrator.

An integrator is op-Amp circuits whose output is proportional to the integral of the input signal.

2. State the applications of an integrator.

Applications of an integrator include waveform generation analog computation and signal processing.

3. What is a differentiator?

A differentiator is an op-Amp circuit whose output is proportional to the rate of change of the input signal.

4. What is the modification to be done in differentiator to work as integrator?

To make a differentiator work as an integrator, interchange the resistor and capacitor in the input and feedback paths.

5. What is the requirement of a lowpassfilter to actasintegrator?

For a low-pass filter to act as an integrator, the input frequency must be much higher than the cutoff frequency.

6. Write the gain equation of the integrator.

The gain equation of an integrator is,

$$A_v = -\frac{1}{j\omega RC}$$

7. What is called lossy integrator?

A lossy integrator is an integrator circuit with a resistor connected in parallel with the feedback capacitor to provide finite DC gain.

Viva Questions:

1. What is an instrumentation amplifier?

An instrumentation amplifier is a special type of differential amplifier used to amplify small differential signals accurately in the presence of noise.

2. What are the features of instrumentation amplifier?

⇒ High input impedance ⇒ Low offset voltage ⇒ High CMRR
⇒ High gain accuracy ⇒ High stability ⇒ Low noise

3. What are the advantages of instrumentation amplifier?

*. Accurate amplification of very small signals.
*. Useful in medical, industrial and measurement systems.

4. Write the expression for output voltage.

$$V_0 = \left(1 + \frac{2R}{R_G}\right)(V_2 - V_1)$$

V_1, V_2 → input voltage
R_G → controls gain.

5. What is the use of transducer in an instrumentation amplifier?

A transducer converts a physical quantity into an electrical signal, which is then amplified by instrumentation amplifier.

6. What is the need for an instrumentation amplifier?

An instrumentation amplifier is needed to amplify small differential signals and reject common mode noise.

7. List out the features of instrumentation amplifier.

⇒ High input impedance ⇒ Low offset voltage
⇒ High CMRR ⇒ Low noise ⇒ High gain accuracy

8. Give some of the practical applications of instrumentation amplifier.

⇒ Biomedical instruments ⇒ Strain-gauge measurement
⇒ Temperature measurements ⇒ Data acquisition systems.

9. Mention some of the linear applications ofop–amps.

⇒ Inverting amplifier ⇒ Voltage follower ⇒ Integrator
⇒ Non-inverting amplifier ⇒ Summing amplifier ⇒ Differentiator.

10. Mention some of the non–linear applications of op-amps.

⇒ comparator ⇒ clamping circuit
⇒ Rectifier ⇒ Log and antilog amplifier.
⇒ Peak detector

Viva Questions:

1. What is a voltage regulator?

A voltage regulator is a circuit that maintains a constant output voltage despite changes in input voltage or load.

2. Give the classification of voltage regulators:

⇒ Linear voltage regulator.

⇒ Switching voltage regulator.

3. What is a linear voltage regulator?

A linear voltage regulator is a circuit that provides a constant voltage output by continuously controlling the pass transistor.

4. What is a switching regulator?

A switching regulator is a circuit that maintains a constant output voltage by rapidly switching a transistor ON and OFF.

5. What are the advantages of IC voltage regulators?

⇒ Small size ⇒ Easy to use.

⇒ Low cost ⇒ Good voltage protection

⇒ High reliability ⇒ Built-in protection

6. Give some examples of monolithic IC voltage regulators:

Examples of monolithic IC voltage regulators are,

IC 7805, IC 7812, IC 7905, IC 7912.

7. What is the purpose of having input and output capacitors in three terminal IC regulators?

The input capacitor reduces input voltage fluctuations, while the output capacitor reduces output ripple and improves stability of the regulator.

8. Give the drawbacks of linear regulators:

⇒ Low efficiency

⇒ High power dissipation

⇒ Requires heat sink

⇒ Limited voltage difference between input and output.

⇒ Not suitable for high-power applications

Viva Questions:

1. Why an LC tank circuit does not produce sustained oscillations. How can this can be overcome?

An LC tank circuit loses energy due to resistance. So oscillations decay, this is overcome by using a feedback amplifier.

2. Differentiate oscillator from amplifier.

An oscillator generates signals without an external input, whereas an amplifier amplifies an external input signal.

3. State Barkhausen criterion for sustained oscillation.

⇒ The loop gain must be unity.

⇒ Total phase shift must be 0° or 360°

4. What will happen to the oscillation if the magnitude of the loop gain is greater than unity?

If the loop gain is greater than unity, the oscillations increase in amplitude and become unstable

5. What is the output of an oscillator if transistor is ideal?

For an ideal transistor, the oscillator produces a pure sinusoidal output with constant amplitude.

6. What is the frequency range of LC oscillators?

The frequency range of LC oscillator is typically 100 kHz to 500 MHz.

7. What is the output frequency expression for colpitts and hartley oscillator?

⇒ Colpitts oscillator : $f = \frac{1}{2\pi\sqrt{LC}}$, where $C = \frac{C_1 C_2}{C_1 + C_2}$

⇒ Hartley oscillator: $f = \frac{1}{2\pi\sqrt{LC}}$, where $L = L_1 + L_2$

8. What are the advantages and disadvantages of Colpitts oscillator?

The colpitts oscillator has good frequency stability and suitable for high frequency applications, but it is difficult tune and not suitable for low-frequency applications.

9. What are the advantages and disadvantages of Hartley oscillator?

The Hartley oscillator offers easy tuning and is suitable for high-frequency applications, but it has lower frequency stability and is affected by inductor losses.

10. List the applications of LC oscillator.

⇒ RF signal generation ⇒ Local oscillator

⇒ Radio transmitters and receivers

⇒ Frequency synthesizers ⇒ communication systems.

Tabulation:

Oscillator	Amplitude (V)	Time Period (S)	Frequency (Hz)
RC Phase Shift Oscillator			
Wien Bridge Oscillator			

Inference:

Viva Questions:

1. What is the frequency of RC phase shift oscillator?

For an RC phase-shift oscillator with three identical RC sections:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

2. What is phase shift oscillator?

A phase shift oscillator is an oscillator that uses an RC network to provide the required phase shift for sustained oscillations.

3. Why RC oscillators cannot generate high frequency oscillations?

RC oscillators cannot generate high-frequency oscillations because stray capacitances and parasitic effects become significant at high frequencies.

4. What are the applications of RC phase shift oscillators?

Applications of RC phase-shift oscillators includ audio signal generation, function generators and low-frequency signal sources.

5. What phase shift does RC phase shift oscillator produce?

An RC phase-shift oscillator produces a total phase shift of 360° (or 0°) for sustained oscillation

6. Why we need a phase shift between input and output signal?

A phase shift is created to provide the required positive feedback and maintain sustained oscillations.

7. What are the advantages and disadvantages of RC phase shift oscillators?

The RC phase-shift oscillator has good frequency stability and is suitable for low-frequency generation, but it has low output power and is not suitable for high-frequency applications.

8. How can we get a maximum phase angle of 90 degrees in RC phase shift oscillator?

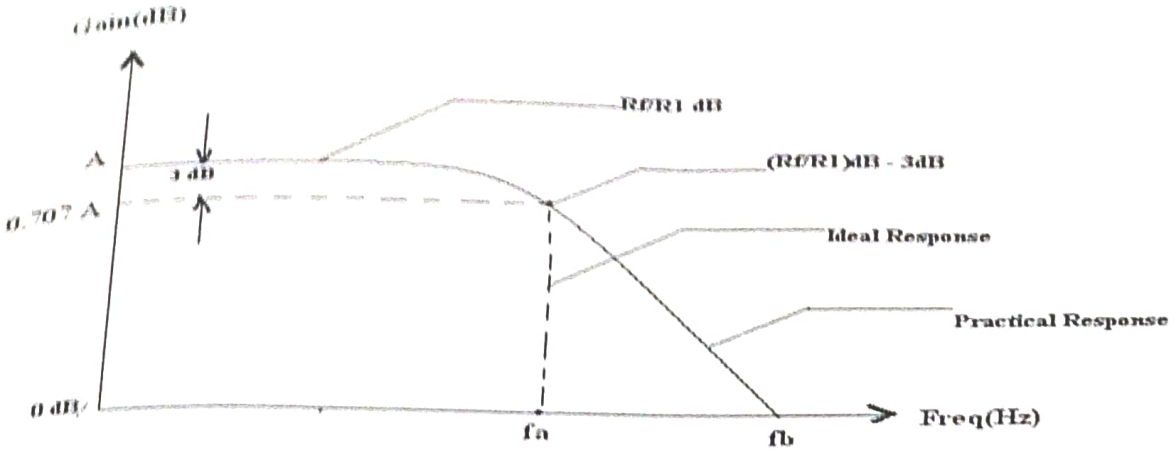
A maximum phase angle of 90° is obtained from each RC section when the capacitive reactance equals the resistance ($X_C = R$)

9. What is the necessary condition for a Wien bridge oscillator circuit to have sustained oscillations?

The necessary condition is that the loop gain must be unity ($A\beta = 1$) and the total phase shift must be 0°.

10. What are the advantages and disadvantages of wein bridge oscillators?

The wein bridge oscillator has good frequency stability, low distortion, and easy tuning, but it is not suitable for high-frequency applications and requires amplitude stabilization.



Inference:

Viva Questions:

1. What is Biasing?

   Biasing is the process of applying DC voltage or current to set the operating point of a transistor.

2. What is CMRR?

   CMRR (common Mode Rejection Ratio) is the ability of an amplifier to reject common-mode signals while amplifying the differential signal

3. What is an operational amplifier?

   An operational amplifier is a high-gain differential amplifier used to amplify and process electrical signals.

4. What is open loop and close loop opamp?

   An open loop op-amp operates without feedback while a closed loop op-amp operates with feedback.

Viva Questions:

1. Define an integrator.

An integrator is an opamp circuit whose output is proportional to the integral of the input signal.

2. State the applications of an integrator.

Application of an integrator include waveform generation, analog computation and signal processing.

3. What is the modification to be done in differentiator to work as integrator?

To convert a differentiator into an integrator, interchange the resistor and capacitor in input and feedback paths.

4. What is Schmitt trigger?

A schmitt trigger is a comparator circuit with hysteresis that converts a noisy or slowly varying input signal into a clean digital output.

5. Why positive feedback is used in comparator?

positive feedback is used in comparator to provide hysteresis and ensure fast, stable switching between output states.

6. What is hysteresis?

Hysteresis is the difference between the upper and lower threshold voltages of a schmitt trigger.

Inference:

DC condition

VBE = VCE =

Square wave response

Visualization with LED

LED turns ON and OFF 5 times in one second

Viva Questions:

1. What is charge?

Electrical charge is a physical property of matter responsible for electrical forces and interactions. Its unit is Coulomb (C).

2. Define current?

Electric current is the rate of flow of electric charge through a conductor. Its unit is Ampere (A)

3. What is voltage?

Voltage is the potential difference between two points that causes electric current to flow. Its unit is Volt (V)

4. Define power.

Electric power is the rate at which electrical energy is consumed or transferred. Its unit is Watt (W)

5. What is network?

An electrical network is an interconnection of electrical components such as resistors, capacitors, inductors and sources.

6. Distinguish between a branch and a node of a circuit.

A branch is a single element or path between two nodes, while node is a junction where two or more branches meet.

7. Distinguish between a mesh and a Loop of a circuit.

A loop is any closed path in a circuit, while a mesh is a closed path that contains no other loops within it.

8. Define active and passive elements.

Active elements can supply energy to a circuit, while passive elements can only absorb or store energy.

9. Define unilateral and bilateral elements.

unilateral elements allow current to flow mainly in one direction, while bilateral elements allow current to flow in both directions.

10. Define linear and non-linear elements.

Linear elements have a proportional relationship between voltage and current, while non-linear elements do ~~not~~ have a proportional relationship between voltage and current.